

Chronic liver disease

Definition: Chronic liver disease (CLD) refers to a long-term pathological process of continuous destruction of liver parenchyma for more than six months and its gradual substitution with fibrous tissue, and characterized by decreased hepatic functions that include the production of clotting factors and other proteins, detoxification of harmful products of metabolism, and excretion of bile.

Continuous process of inflammation, destruction, and regeneration of liver parenchyma is leading to fibrosis and then cirrhosis which is a final stage of chronic liver disease that results in disruption of liver architecture, the formation of widespread nodules, vascular reorganization, neo-angiogenesis, and deposition of an extracellular matrix.

Etiology

The spectrum of etiologies is broad for chronic liver disease, which includes toxins, alcohol abuse for a prolonged time, drugs, infection, autoimmune diseases, genetic and metabolic disorders.

Chronic liver disease is an extremely common clinical condition, and the focus is done on the common etiologies, clinical manifestations, and management.

The following are the most common etiologies:

Alcoholic Liver Diseases : Includes alcoholic fatty liver with or without hepatitis, alcohol hepatitis (reversible because of acute ingestion) to cirrhosis (irreversible).

Non-alcoholic Fatty Liver Disease (NAFLD/NASH): Usually has an association with metabolic syndrome (obesity, hyperlipidemia, and diabetes mellitus). Some of these patients develop non-alcoholic steatohepatitis, which leads to fibrosis of the liver. All the risk factors of metabolic syndrome can aggravate the disease process.

Chronic Viral Hepatitis: Chronic hepatitis C, B and D infections are common causes of chronic liver disease. There are various genotypes of hepatitis C. In Europe and North America, genotype 1a and 1b are more prevalent, while in Southeast Asia, genotype 3 is more common. A molecular epidemiological study revealed a high prevalence of HCV genotype 4, subtype 4a among Egyptian patients living in Sharkia governorate, Egypt. Chronic hepatitis C, if not treated, may lead to hepatocellular carcinoma.

Genetic Causes

- **Alpha-1 antitrypsin deficiency:** This is a common genetic cause of CLD among children.
- **Hereditary hemochromatosis:** It is an autosomal recessive disorder of iron absorption due to a mutation of the HFE gene that regulates the iron absorption from the intestine, excessive iron is absorbed from the gastrointestinal tract resulting in a pathological increase in total body iron (such as ferritin). This is leading to more generation of hydroxyl free radicals, which in turn causes organ fibrosis.

Wilson disease: Autosomal recessive disorder due to deficiency of ceruloplasmin (copper carrying protein) leading to excessive amounts of copper accumulation in the

body, especially in the brain, liver and eyes. The signs and symptoms of Wilson disease usually include a combination of liver disease, neurological and psychiatric problems.

Nervous system or psychiatric problems are often the initial features with clumsiness, tremors, difficulty walking, speech problems, impaired thinking ability, mood disturbance as depression and anxiety.

The signs and symptoms of liver disease include jaundice, fatigue, loss of appetite, and may be ascites.

Copper deposits in the cornea forming a ring, called Kayser-Fleischer ring.

Autoimmune Causes

Autoimmune hepatitis is a rare disease in which there is the destruction of liver parenchyma by autoantibodies. Most of the patients who present with this disease have already developed cirrhosis. Females are more commonly affected than males.

- **Primary biliary cirrhosis (PBC):** This is an autoimmune and progressive disease of the liver, which is the destruction of intrahepatic biliary channels and portal inflammation and scarring. It leads to cholestatic jaundice and fibrosis of liver parenchyma. PBC is more common in middle-aged women. Alkaline phosphatase levels increase in PBC.
- **Primary Sclerosing Cholangitis (PSC):** commonly associated with ulcerative colitis. This condition is characterized by a decrease in the size of intrahepatic and extrahepatic bile ducts due to inflammation and fibrosis.

- **Autoimmune hepatitis (AIH):** This is a form of chronic inflammatory hepatitis, more common in women than men, and is characterized by elevated autoantibodies such as antinuclear antibodies, anti-smooth muscle antibodies, and hypergammaglobulinemia.

Other Causes of Chronic Liver Disease

- Drugs - amiodarone, isoniazid, methotrexate, phenytoin, nitrofurantoin.
- Vascular. Budd-Chiari syndrome.
- Idiopathic/cryptogenic, around 15%

Epidemiology

Chronic liver disease is one of the frequent causes of death, especially in the developing world. The increasing prevalence of chronic liver disease has been noted in recent times. The majority of chronic liver diseases in the developed world include alcoholic liver disease, chronic viral hepatitis, including hepatitis B and C, non-alcoholic fatty liver disease (NAFLD), and hemochromatosis.

Pathophysiology

Hepatic fibrosis is the deposition of extracellular matrix (ECM) in response to chronic liver injury by any etiology.

The underlying mechanism of fibrosis and cirrhosis at a cellular level is the recruitment of stellate cells and fibroblasts that cause fibrosis, while parenchymal regeneration relies on hepatic stem cells.

The common pathway is initiated by hepatic stellate cells (HSC), which usually are vitamin A storing dormant cells found in space between sinusoids and hepatocytes. In response to chronic liver injury, HSC gets activated into proliferative fibrogenic myofibroblast and upregulates the expression of inflammatory receptors such as chemokine receptors, ICAM-1, and other inflammatory mediators by releasing chemokines and other leukocyte chemoattractants. This pro-inflammatory phase or initiation phase also changes gene and phenotypic expression of the liver cells, making them more responsive to these inflammatory cytokines, and perpetuation of activated HSC cells resulting in accumulation of ECM and progressive fibrosis.

This activity reviews the evaluation and treatment of chronic liver disease and highlights the role of the healthcare team in evaluating and treating patients with this condition.

Clinical Manifestations

Signs and symptoms of CLD can be nonspecific, such as fatigue, anorexia, weight loss, or depend upon the complication that the patient has developed.

Signs and symptoms of significant complications as portal hypertension (esophageal varices, ascites), hepatocellular insufficiency (e.g., jaundice, hepatic encephalopathy), and hepatocellular carcinoma. Decompensated chronic liver disease can present with one of the following complications.

(A) Portal Hypertension: PH is a result of resistance to portal blood flow because of cirrhotic and noncirrhotic etiology. A portal venous pressure above 7 mmHg is considered portal

hypertension; however, clinical features or complications do not develop until portal pressure is higher than 12 mmHg. Portal hypertension causes can divide into prehepatic (e.g., portal vein thrombosis), hepatic (e.g., cirrhosis), and post hepatic (e.g., Budd Chiari syndrome). Cirrhosis and hepatic schistosomiasis remain the most common cause of portal hypertension. The following are the consequences of long-standing portal hypertension.

- Esophageal or gastric varices: It presents with melena or upper GI bleeding and is the most common life-threatening complication of CLD.
- Caput medusa or rectal hemorrhoids
- Ascites: It is an accumulation of fluid in the peritoneal cavity because of raised portal pressure (increased hydrostatic pressure), decreased albumin (reduced oncotic pressure), and splanchnic vasodilation (due to the release of nitric oxide). Most of the patients develop ascites in the later stages of cirrhosis. Clinical findings in such patients are abdominal distension, shifting dullness, and a fluid wave. Tense ascites can lead to shortness of breath or early satiety.

(B) Hepatic Encephalopathy:

This is a neuropsychiatric syndrome caused by hepatic dysfunction due to failure of detoxification of harmful products of metabolism, e.g., ammonia leading to an increased level of ammonia, which impair consciousness.

There are different grades of hepatic encephalopathy.

- **Grade 0/Minimal:** Subclinical, normal mental status with minimal changes in memory, coordination, intellectual function, concentration.
- **Grade 1:** Trivial lack of awareness, euphoria or anxiety, shortened attention span, impairment of addition or subtraction, altered sleep rhythm.
- **Grade 2:** Lethargy or apathy, disorientation to time, personality change, inappropriate behavior, dyspraxia, asterixis.
- **Grade 3:** Somnolence to semi-stupor, responsive to stimuli, confused, gross disorientation, bizarre behavior.
- **Grade 4:** Coma

(C) Jaundice

Jaundice is clinically visible when total bilirubin is greater than 2 mg/dl. As in chronic liver disease, there is the destruction of liver parenchyma, and it does not conjugate bilirubin, which deposits in various tissues of the body. There is pruritus because of the accumulation of bile salts.

(D) Spontaneous Bacterial Peritonitis (SBP)

It is one of the acute and painful complications of chronic liver disease. Bacteria (*E. coli*, *Klebsiella*, *Streptococcus pneumonia*) seep through the gastrointestinal tract and infect the ascitic fluid. The infection spread through the fluid to the peritoneal membrane, causing inflammation. SBP presents with fever, generalized abdominal pain, tenderness, and absent bowel sounds.

(E) Hyperestrinism

In chronic liver disease, the catabolism of estrogen becomes impaired, resulting in excess estrogen in the body. This manifests as palmar erythema, spider angiomas (dilated cutaneous arterioles with a central red spot and red extensions that radiate outward like a spider's web in the territory of SVC), gynecomastia (enlarged tender subareolar tissue), and testicular atrophy.

(F) Hepatorenal Syndrome (HRS)

Hepatorenal syndrome is a functional renal failure as kidneys are normal, where there is a gradual loss of renal function. It is a diagnosis of exclusion. Vasoconstrictors are released in CLD, which is responsible for the narrowing of renal vessels. The following criteria have been described:

- Chronic liver disease with portal hypertension or advanced liver failure
- Continuous rising in creatinine, usually more than 0.3 mg/dl within 48 hours or doubling from baseline within seven days.
- Oliguria with the absence or minimal proteinuria
- Urine sodium less than 10 meq/L
- Failure to improve with volume expansion and stopping the diuretics.
- Absence of shock
- No recent use of the nephrotoxic drug
- Absence of renal parenchymal disease

(G) Coagulopathy

The liver produces clotting factors, so the patients with CLD have coagulopathies and manifest or contribute to easy bruising and bleeding per gastrointestinal tracts. Hence, PT/INR (intrinsic pathway) and APTT (extrinsic pathways) are prolonged.

Diagnosis:

The diagnosis of chronic liver disease depends upon the etiology and complications of the disease. An outline regarding the diagnosis for various CLD appears below.

- Viral hepatitis B, D and C: Serology, PCR (quantitative or qualitative) with genotype
- Alcoholic liver disease: Elevated levels of AST>ALT with a history of chronic alcohol intake. Usually, AST: ALT ratio is 2 to 1 in alcoholic liver disease.
- Hemochromatosis: Raised serum iron, ferritin, decreased TIBC, and liver biopsy. Genetic testing can detect a mutation in the HFE gene, especially C282Y.
- Wilson disease: Raised urine copper, decreased serum ceruloplasmin, and liver biopsy. Genetic testing for the ATP7B gene.
- Non-alcoholic fatty liver disease: Diagnosis of exclusion and ALT>AST. Ultrasonography of the liver is informative.
- Autoimmune hepatitis: Raised ANA, ASMA, and LKM-1
- Alpha 1 antitrypsin deficiency: Decreased levels of alpha one antitrypsin

- Primary biliary cirrhosis: Markedly raised alkaline phosphatase levels with an antimitochondrial antibody
- Budd-Chiari and veno-occlusive disease: CBC, clotting profile, and imaging studies like ultrasound doppler or computed tomography with contrast study

Laboratory Findings

In chronic liver disease, there is inflammation and destruction of hepatocytes that leads to the release of alanine aminotransferase (ALT) and aspartate aminotransferase (AST). but normal levels of these markers do not rule out cirrhosis. As in compensated CLD LFTs may be normal in cirrhosis.

Other parameters (ALP and GGT) of LFTs also appear elevated in cholestatic conditions like PBC. Bilirubin (unconjugated > conjugated) levels are elevated in jaundice. There is reduced production of clotting factors leading to raised PT/INR and APTT.

Because of hepatocellular insufficiency in cirrhosis, albumin level drops, and ammonia raises, causing ascites and hepatic encephalopathy.

In the case of ascites, diagnostic paracentesis is an investigation of choice to see whether ascites are because of raised portal pressure or another cause.

Radiologic Investigations

- **Ultrasound abdomen** to detect the size, echogenicity, nodularity of the liver, thereby diagnosing liver cirrhosis, measurement of portal vein diameter which increases in

portal HTN and assessment of a clot in the hepatic vein (Budd-Chari) and portal vein in portal vein thrombosis.

- **Triphasic CT** scan can show a lesion in the liver or obstruction of biliary channels in a more precise way and is the test of choice in diagnosing hepatocellular carcinoma.
- **Transient elastography** (TE) detects early stages of cirrhosis. It can also detect cardiovascular damage in patients with NAFLD.
- **Wedge hepatic venous pressure** measures portal venous pressure in CLD.
- **Doppler scan** can help in diagnosing Budd-Chiari and portal vein thrombosis.
- **EEG** shows delta waves in hepatic encephalopathy.
- **An upper endoscopy** can diagnose and treat esophageal varices.
- **Liver biopsy** can confirm the diagnosis of chronic liver disease.

There are various scoring systems used to assess the severity of chronic liver disease.

1. **The Child-Pugh score** uses ascites, bilirubin, albumin, PT, and encephalopathy.

- Class A (score 5-6): well-compensated disease.
- Class B (score 7-9): functional compromise.
- Class C (score 10-15): decompensated disease.

2. **MELD** (model of end-stage liver disease) uses bilirubin, serum creatinine, and INR. The MELD score was initially used to predict mortality within three months after the TIPS procedure and now used to prioritize patients for receipt of a liver transplant.

Complications of chronic liver disease:

- Variceal bleeding
- Ascites and spontaneous bacterial peritonitis (SBP)
- Hepatic encephalopathy
- Hepatorenal syndrome
- Hepatopulmonary syndrome
- Hepatocellular carcinoma

Prophylaxis: Recommendations for Preventive Care are:-

- Avoid various types of alcohol.
- Regular screening for hepatitis B and hepatitis C
- Vaccination against hepatitis A and B
- Avoid iron supplementation unless there is an iron deficiency.
- Avoid hepatotoxic drugs.
- Maintain a good lipid profile to avoid metabolic syndrome and NAFLD.

- To prevent complications, patients with chronic liver disease will need surveillance. These patients should undergo routine monitoring of their CBC and prothrombin time at least 3 to 4 times per year.
- Routine diagnostic endoscopy should be performed for asymptomatic esophageal varices, and a follow-up endoscopy should be done in 2 years if varices are not present. Treatment with a nonselective beta-blocker to reduce heart rate by 25% in patients with esophageal varices, offers primary prophylaxis with variceal bleeding. Patients with large esophageal varices should have prophylactic endoscopic variceal banding to avoid the risk of rupture and bleeding.
- Surveillance of patients with cirrhosis for HCC with ultrasonography every six months. A triphasic CT scan or a dynamic MRI scan should be done to rule out hepatocellular carcinoma in patients with a liver nodule on the ultrasound.

Treatment:

General treatment:

The treatment goal is to stop the progression of the disease and complications and treating the underlying cause, Portal hypertension management, and specific treatments for individual disease.

Esophageal varices related bleeding treatment includes aggressive fluid resuscitation, vasopressors (octreotide, terlipressin), and endoscopy. Endoscopic band ligation and

injection sclerotherapy are the usual modalities to treat variceal bleed in an emergency. In some patients, early transjugular intrahepatic portosystemic shunt (TIPS) can increase the survival rate. Propranolol is used for primary and secondary prophylaxis for esophageal varices.

Ascites: Diuretics (furosemide, spironolactone) and sodium restriction are essential treatment options for ascites. For tense ascites, therapeutic paracentesis is done. Albumin infusion can also be considered. Initially,

SBP: broad-spectrum antibiotics are the treatment of choice for SBP and then specific antibiotics after culture.

Hepatic encephalopathy - The basic principle of treatment is to address the precipitating factors. Patients with hepatic encephalopathy usually improve with precipitating cause correction along with rifaximin and lactulose. Lactulose acts by converting ammonia to ammonium ions and decreases its absorption from the gastrointestinal tract. Lactulose also relieves constipation through its osmotic effect, which further helps to ease the symptoms of hepatic encephalopathy. Rifaximin is used to decrease ammonia production by gut flora.

Hepatorenal syndrome based on severity is divided into two categories. HRS 1 is more severe compared to HRS 2 (less severe). The primary goal is to correct underlying cause correction to reverse acute kidney injury. Treatment modalities depend on the severity and location of the patient. Treatment modalities include norepinephrine or terlipressin with albumin infusion or midodrine, octreotide with albumin infusion. TIPS procedure in some patients

can help and liver transplantation the only definite treatment in the patient who fails to respond to all other treatments.

Hepatocellular carcinoma (HCC) - Treatment is based on the Barcelona clinic liver cancer staging system in the management of HCC:

- Initial stage (single HCC lesion): Transplantation, surgical resection and radiofrequency or microwave ablation.
- Intermediate stage: Transarterial chemoembolization and radio-embolization.
- Metastatic disease: Sorafenib

Specific Treatment

Viral Hepatitis

- Continuous viral suppression with nucleoside and nucleotide analogs
- Direct-acting antivirals achieving HCV eradication

Alcoholic liver disease: Alcohol abstinence

Non-alcoholic fatty liver disease: Treatment of metabolic syndrome components

Autoimmune hepatitis: Corticosteroids and other immunosuppressive drugs

Hereditary hemochromatosis: Phlebotomy, iron-chelators

Copper overload (Wilson disease): Copper chelators

Alpha-1-antitrypsin deficiency: Transplant

Drugs and toxins: Identify and stop the factor

Primary biliary cholangitis (PBC): Ursodeoxycholic acid (UDCA)

Primary sclerosing cholangitis: Transplant

Budd-Chiari syndrome: Anticoagulation, thrombolysis or angioplasty with or without stenting, TIPS, or liver transplant

Prognosis

Compensated chronic liver disease (who has not developed significant complications) usually carries a better prognosis than decompensated liver cirrhosis.

Decompensated liver cirrhosis patients (who have developed variceal bleeding, ascites, HCC, SBP, and hepatorenal syndrome) have a poor prognosis. Mean survival of about six months if Child-Pugh score of 12 or greater or MELD score of 21 or higher.